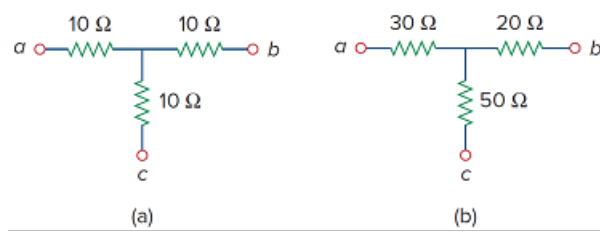


Problem

Convert the circuits in Fig. 2.112 from Y to Δ .

Figure 2.112:



Step-by-step solution

Step 1 of 7

(a)

Draw the circuit diagram.

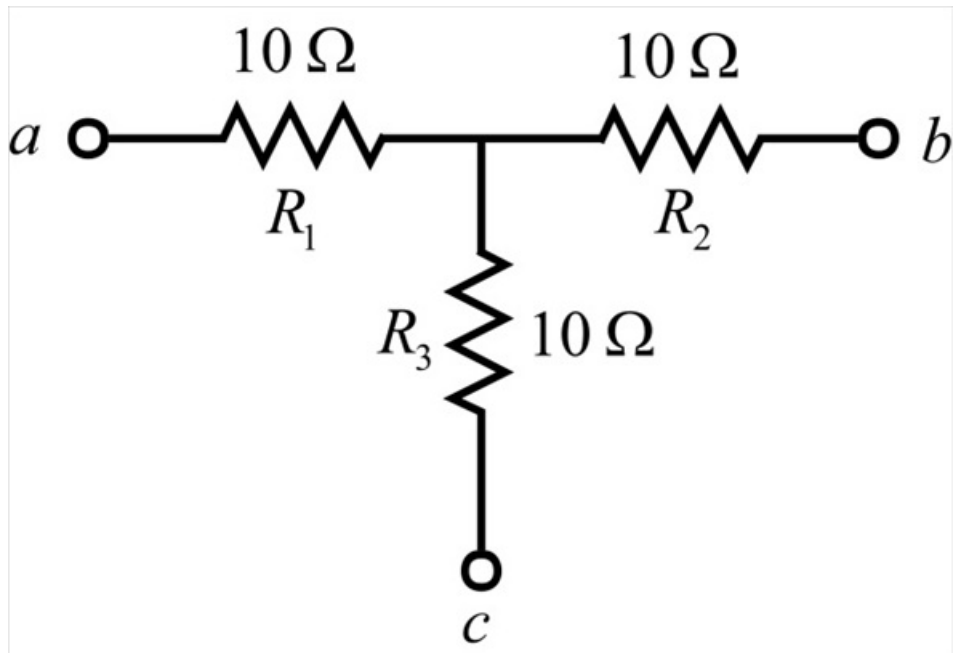


Figure 1

Step 2 of 7

Consider the following data from the circuit shown in Figure 1.

The values of resistances connected in Y network are,

$$R_1 = 10 \, \Omega$$

$$R_2 = 10 \, \Omega$$

$$R_3 = 10 \, \Omega$$

Step 3 of 7

Determine the equivalent resistances connected in delta network.

$$\begin{aligned}
 R_a &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} \\
 &= \frac{(10)(10) + (10)(10) + (10)(10)}{10} \\
 &= \frac{100 + 100 + 100}{10} \\
 &= \frac{300}{10} \\
 &= 30 \, \Omega
 \end{aligned}$$

$$\begin{aligned}
 R_b &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} \\
 &= \frac{(10)(10) + (10)(10) + (10)(10)}{10} \\
 &= \frac{100 + 100 + 100}{10} \\
 &= \frac{300}{10} \\
 &= 30 \, \Omega
 \end{aligned}$$

$$\begin{aligned}
 R_c &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} \\
 &= \frac{(10)(10) + (10)(10) + (10)(10)}{10} \\
 &= \frac{100 + 100 + 100}{10} \\
 &= \frac{300}{10} \\
 &= 30 \, \Omega
 \end{aligned}$$

Step 4 of 7

Now, draw the equivalent delta connected network.

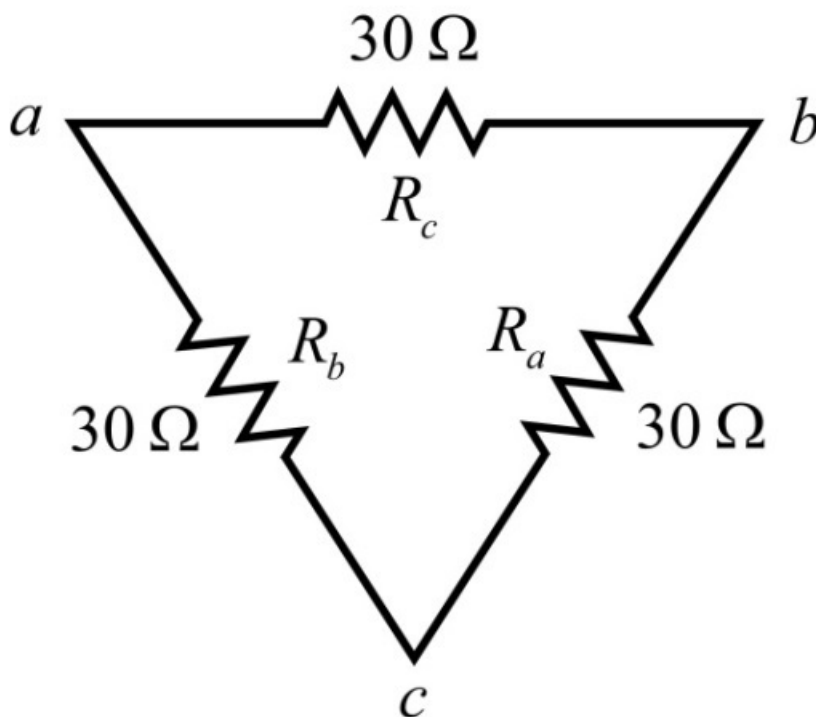


Figure 2

Step 5 of 7

(b)

Draw the circuit diagram.

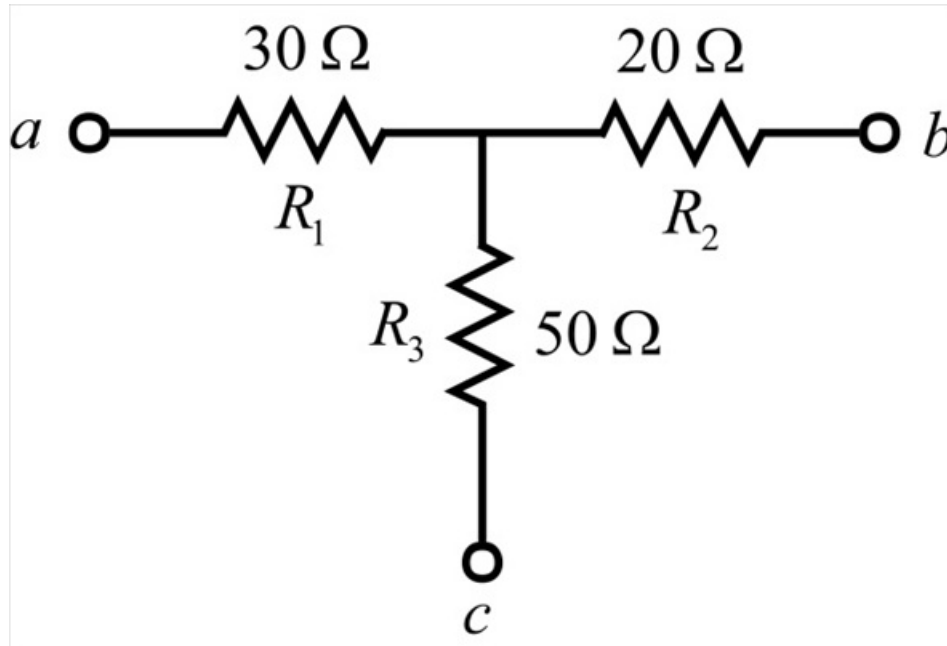


Figure 3

Consider the following data from the circuit shown in Figure 3.

The values of resistances connected in Y network are,

$$R_1 = 30\ \Omega$$

$$R_2 = 20\ \Omega$$

$$R_3 = 50\ \Omega$$

Determine the equivalent resistances connected in delta network.

$$\begin{aligned}
 R_a &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} \\
 &= \frac{(30)(20) + (20)(50) + (50)(30)}{30} \\
 &= \frac{600 + 1000 + 1500}{30} \\
 &= \frac{3100}{30} \\
 &= 103.3 \, \Omega
 \end{aligned}$$

$$\begin{aligned}
 R_b &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} \\
 &= \frac{(30)(20) + (20)(50) + (50)(30)}{20} \\
 &= \frac{600 + 1000 + 1500}{20} \\
 &= \frac{3100}{20} \\
 &= 155 \, \Omega
 \end{aligned}$$

$$\begin{aligned}
 R_c &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} \\
 &= \frac{(30)(20) + (20)(50) + (50)(30)}{50} \\
 &= \frac{600 + 1000 + 1500}{50} \\
 &= \frac{3100}{50} \\
 &= 62 \, \Omega
 \end{aligned}$$

Step 7 of 7

Now, draw the equivalent delta connected network.

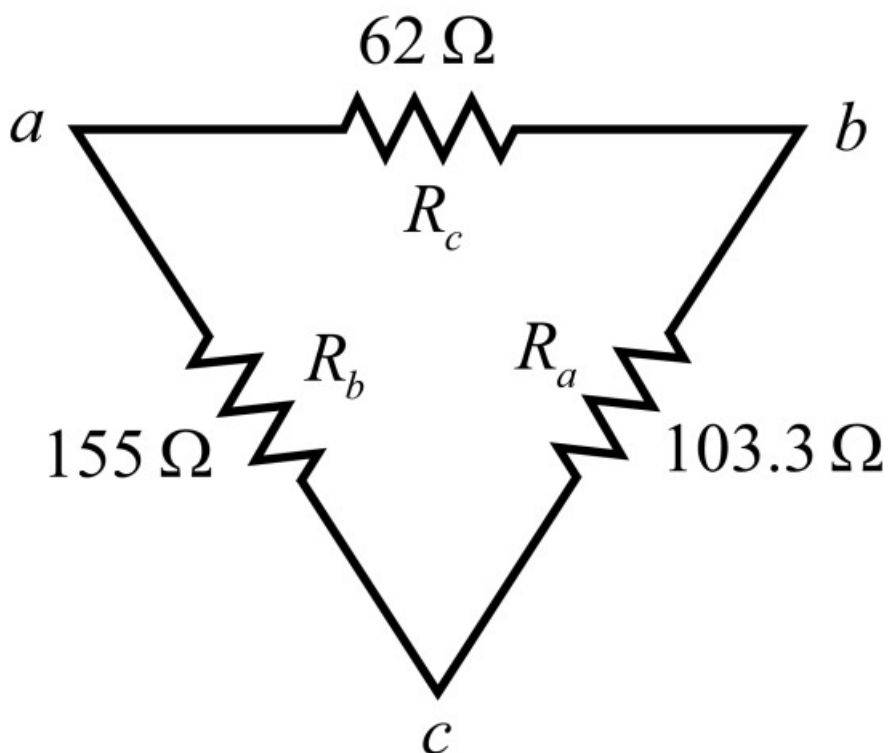


Figure 4